CONFOEDERATIO,
RESEARCH DIVISION (CRD).

DEMOGRAPHIC AND SETTLEMENT HISTORY OF CRIMEA (10000BC - 1400AD).

Natural Sub-Regional Boundaries, Archaeological Synthesis and Population Estimates across the Tauric Peninsula.

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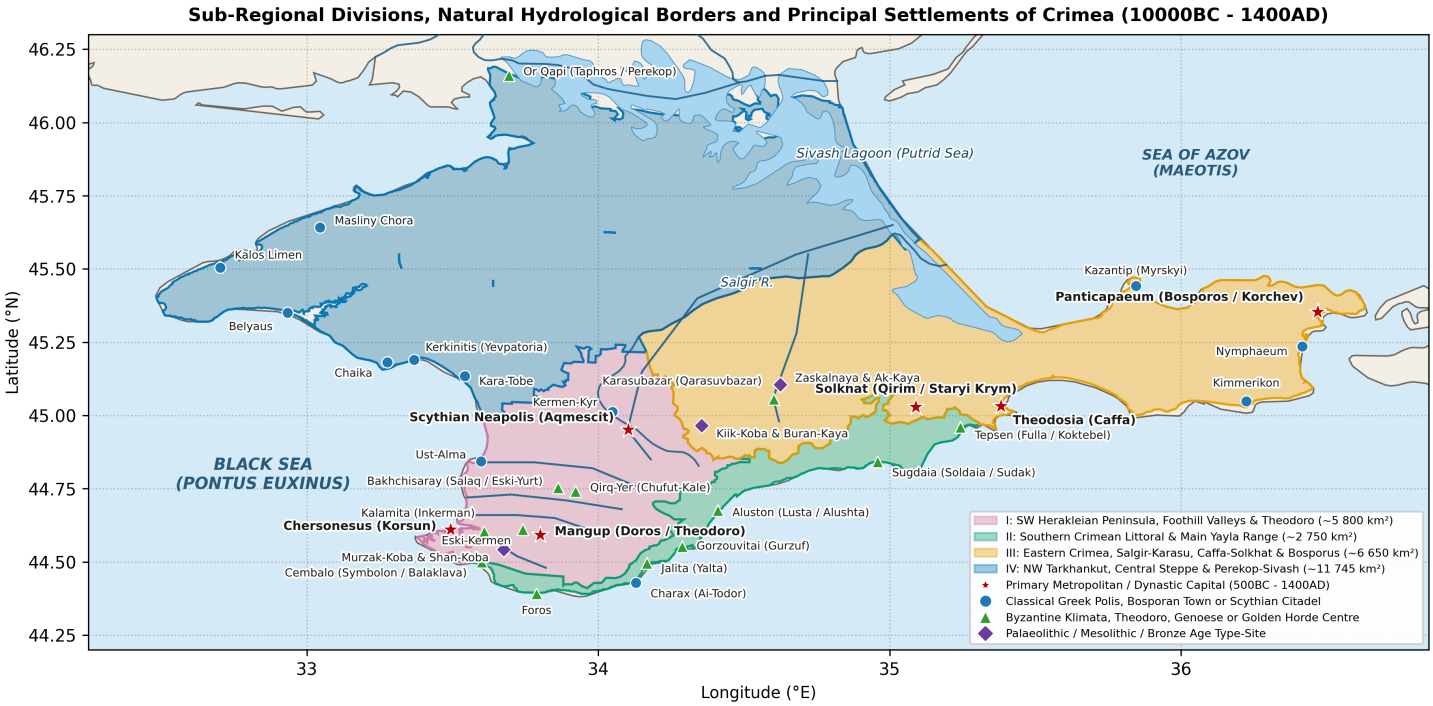


Figure 1. Natural sub-regional divisions, hydrological basins and principal prehistoric, classical and medieval settlements of Crimea (10000BC - 1400AD).

Regional Subdivisions and Geographic Scope

This literature review compiles the demographic and settlement history of the Crimean Peninsula (*Taurica*, ~26 945 km^2 land area; 505 5-arcminute HYDE grid cells covering 30 488 km^2 including coastal cells) from the Final Pleistocene and Early Holocene (10000BC) to the late Golden Horde, Genoese and Theodorite benchmark of 1400AD (post-1400AD developments under the independent Crimean Khanate, Ottoman Eyâlet of Kefe and Imperial/Modern eras require separate regime handling). Bounded by the Black Sea (*Pontus Euxinus*) to the west and south, the shallow Sea of Azov (*Palus Maeotis*) and the hypersaline Sivash lagoon system to the north-east and the Kerch Strait (*Kimmerian Bosporus*) to the east, Crimea connects to the Pontic steppe via the 7 km-wide Perekop Isthmus (*Taphros / Or Qapi*) and the narrow Arabat Spit. Internally, the peninsula divides along natural topographic and hydrological boundaries into three primary ecological belts: the northern and central dry-steppe plain (~72% of land area), the three parallel cuestas of the Crimean Mountains (the Outer, Inner and Main *Yayla* ranges, rising to 1 545 m at Roman-Kosh) dissected by the Chorna, Belbek, Kacha, Alma, Salgir and Biyuk-Karasu

valleys, and the sheltered sub-Mediterranean southern littoral south-east of the Main Yayla crest.

To align the dasymetric areal mask (`crimea_areal_mask.png`) and vector cartography strictly with natural topographic, hydrological and historical boundaries, this synthesis partitions Crimea into four physiographic sub-regions:

1. **Sub-region I: South-Western Herakleian Peninsula, Premontane Valleys and Cave-Town Principality of Theodoro** (~5 800 km^2 land area; 75 grid cells): Bounded on the south-east by the crest of the Western Main Yayla Ridge (north of the Baidar-Balaklava coastal scarp) and encompassing the Herakleian Peninsula (*Chersonesus Taurica* / *Korsun*), the Chorna, Belbek, Kacha, Alma and Upper Salgir foothill valleys, the cuesta table-mountains (*mesas*) of the Inner Range (Mangup-Doros, Eski-Kermen, Chufut-Kale / *Qirq-Yer*, Bakla, Tepe-Kermen, Kachi-Kalyon, Kalamita) and the Bakhchisaray and Simferopol (*Scythian Neapolis*) basins.
2. **Sub-region II: Southern Crimean Littoral and Main Yayla Mountain Range** (~2 750 km^2 land area; 33 grid cells): Bounded naturally on the north-west by the watershed crest of the Main Yayla Range (Ai-Petri, Yalta, Babugan, Chatyr-Dag, Demerdzhi and Karabi *Yaylas*), encompassing the narrow sub-Mediterranean coastal amphitheatres from Cembalo (Symbolon / Balaklava) and Foros eastward past Simeiz, Alupka, Charax, Oreanda, Jalita (Yalta), Gorzouvitai (Gurzuf), Partenit, Lambat and Aluston (*Lusta* / Alushta) to Sugdaia (*Soldaia* / Sudak). Historically administered as the Byzantine *Parathalassia* (*Klimata*) and the Genoese *Capitanate of Gotthia*.
3. **Sub-region III: Eastern Crimea, Salgir-Biyuk-Karasu Valley, Caffa-Solkhat Basin and Kerch Bosporus** (~6 650 km^2 land area; 162 grid cells): Bounded on the north-west by the diagonal course of the Salgir River channel toward the Sivash lagoon, encompassing the Biyuk-Karasu valley (Karasubazar), the Golden Horde regional capital of Solkhat (*Qirim* / Staryi Krym), the Feodosia (*Theodosia* / *Caffa*) bay and Karadag-Koktebel (*Tepsen*) littoral, and the European heartland of the Bosporan Kingdom on the Kerch Peninsula (Panticapaeum / *Bosporos* / *Korchev*, Nymphaeum, Myrmekion, Tyritake, Kimmerikon, Kytaia, Akra and the fortified rural *chora* behind the Uzunlarsky and Tyritake ramparts).
4. **Sub-region IV: North-Western Tarkhankut Peninsula, Central Crimean Steppe and Perekop-Sivash Isthmus** (~11 745 km^2 land area; 235 grid cells): Bounded on the south-east by the Salgir River channel and on the south by the Outer Cuesta foothills, encompassing the limestone coastline of the Tarkhankut Peninsula and the North-Western Chersonesean *chora* (Kerkinitis, Kalos Limen, Chaika, Belyaus, Panskoye I, Masliny, Kara-Tobe, Kulchuk, Tarpanchi), the central Crimean dry steppe and the northern fortified gateway at Perekop (*Taphros* / *Or Qapi*) and the Sivash lagoons.

Sub-Region	Area (% Share)	Principal Urban & Archaeological Centres	Key Benchmark Populations
I. SW Herakleian, Foothill Valleys & Theodoro	5 800 km^2 (21,5%)	Chersonesus (Korsun), Scythian Neapolis, Mangup (Theodoro), Qirq-Yer (Chufut-Kale), Eski-Kermen, Kalamita	17,5k (1000BC); 78,0k (200AD); 74,0k (900AD); 78,0k (1300AD); 62,0k (1400AD)
II. Southern Crimean Littoral & Main Yayla	2 750 km^2 (10,2%)	Cembalo (Symbolon), Foros, Charax, Jalita (Yalta), Gorzouvitai (Gurzuf), Partenit, Aluston, Sugdaia (Soldaia)	7,0k (1000BC); 19,0k (200AD); 36,0k (900AD); 46,0k (1300AD); 36,0k (1400AD)

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Sub-Region	Area (% Share)	Principal Urban & Archaeological Centres	Key Benchmark Populations
III. Eastern Crimea, Salgir-Karasu, Caffa-Solkhat & Bosporus	6 650 km^2 (24,7%)	Panticapaeum (Bosporos/ Korchev), Theodosia (Caffa), Solkhat (Qirim), Karasubazar, Nymphaeum, Ilurat, Tepsen	13,0k (1000BC); 94,0k (200AD); 77,0k (900AD); 96,0k (1300AD); 65,0k (1400AD)
IV. NW Tarkhankut, Central Steppe & Perekop-Sivash	11 745 km^2 (43,6%)	Kerkinitis, Kalos Limen, Belyaus, Panskoye I, Masliny, Chaika, Kara-Tobe, Kulchuk, Or Qapi (Taphros/Perekop)	10,5k (1000BC); 34,0k (200AD); 28,0k (900AD); 30,0k (1300AD); 22,0k (1400AD)
Total Crimean Peninsula (crimea, 10000BC - 1400AD)	26 945 km^2 (100,0%)	171 Georeferenced Pre-1400AD Settlements in Stadester Gazetteer	48,0k (1000BC); 225,0k (200AD); 215,0k (900AD); 250,0k (1300AD); 185,0k (1400AD)

Table 1. Natural Physiographic and Historical Sub-Regions of Crimea (26 945 km^2, 10000BC - 1400AD).

Prehistoric Foragers, Pastoralists and Early Iron Age Tauri (10000BC - 1000BC)

Terminal Pleistocene, Shan-Koba / Murzak-Koba Mesolithic and Neolithic (10000BC - 5000BC)

During the Late Glacial and Early Holocene transition (10000BC - 8000BC), when the Black Sea stood 60 to 80 metres below its present level as the freshwater Neoeuxine lake, human settlement in Crimea concentrated in the sheltered limestone rock-shelters (*grottoes*) of the Inner Cuesta and Chornaya-Belbek-Kacha-Zuya river gorges [1][2]. Continuity from Upper Palaeolithic sites such as Buran-Kaya III (on the Burulcha River), Siuren I (on the Belbek), Kabazi II, Zaskalnaya and Kiik-Koba gave rise to two sequential Mesolithic archaeological complexes documented by G. A. Bonch-Osmolovsky, S. N. Bibikov and D. Ya. Telegin [3][4]:

- **The Shan-Koba Culture** (Final Pleistocene to Early Preboreal, c. 10000-7500BC): Characterised by lunate geometric microliths and intensive hunting of red deer, saiga antelope, wild boar and forest-steppe equids across rock-shelters at Shan-Koba, Fatma-Koba Layer 6, Grot Skalisty (on the Bodrak), Zamil-Koba I and Buran-Kaya III [3]. Across the peninsula's river canyons and expanded coastal plain, seasonal foraging bands (35 - 80 persons per multi-family rock-shelter aggregation) supported an estimated regional population of 1,2k (1 200) at 10000BC, rising to 2,1k (2 100) by 8000BC as Boreal warming expanded oak-pistachio woodlands across the foothill valleys.
- **The Murzak-Koba (Tardigravettian / Tardenoisian) Culture** (7500-5500BC) and **Crimean Neolithic** (6000-5000BC): Excavations at Murzak-Koba (in the Chornaya gorge, famous for its double Mesolithic burial showing ritual finger-phalange amputation), Fatma-Koba, Laspi VII on the southern coast, Tash-Air I in the Kacha canyon, Kaya-Arasy near Bakhchisaray, At-Bash on the Ai-Petri *Yayla* and Frontovoye I on the Kerch Peninsula demonstrate a shift toward trapeziform microliths, *Helix* land-snail harvesting, coastal fishing and incipient swine and ovicaprid domestication [1][3][4]. By 6000BC, Crimean population reached 3,5k (3 500), nearly doubling to 6,5k (6 500) at 5000BC with the establishment of permanent Neolithic open air villages at Tash-Air I, Kaya-Arasy, Alekseevka on the Tarkhankut Peninsula and Frontovoye.

Eneolithic, Kemi-Oba Megalithic Culture and the Bronze Age - Tauri Transition (4000BC - 1000BC)

Between 4000BC and 2000BC, Crimea became a contact zone between Pontic steppe kurgan pastoralists (Sredny Stog, Yamnaya and Catacomb cultures) and North Caucasian metalworking traditions [5][6]. During the Early and Middle Bronze Age (3200-2200BC), the indigenous **Kemi-Oba Culture** (named after the kurgan sanctuary near Bilohirsk excavated by A. A. Shchepinsky) occupied the premontane valleys from Sevastopol to the Salgir and Biyuk-Karasu basins. Kemi-Oba communities constructed timber-and-stone cist graves painted with polychrome ochre geometric friezes, erected anthropomorphic stone stelae (*Bakhchi-Eli*, *Kazanki*) and practised mixed valley-bottom agriculture and transhumant *Yayla* grazing [5][6]. Combining Kemi-Oba foothill settlements (Sub-region I: 7,50k in 3000BC, 12,0k in 2000BC) with Yamnaya and Catacomb kurgan groups across the Kerch Peninsula (Kamenka-Bosporan, Planerskoye) and Tarkhankut-Sivash steppes, total Crimean population rose from 11,0k at 4000BC to 22,0k at 3000BC and 35,0k at 2000BC.

During the Late Bronze Age and Early Iron Age transition (1500-800BC), Sabatinovka and Belozerka agricultural villages in the steppe coexisted with the **Kizil-Koba Culture** of the Crimean Mountains and foothills, universally identified by Kh. I. Kris and V. A. Kolotukhin with the historical **Tauri** (*Tauroi*) described by Herodotus (*Histories* IV.99-103) and Euripides [7][8][9]. Anchored by the 12 ha fortified promontory settlement of Uch-Bash overlooking the Chornaya mouth near Inkerman (0,9k - 1,2k inhabitants), the type-site of Kizil-Koba (*Krasnaya Peshchera*) in the Upper Salgir valley, Druzhnoye, Tash-Dzhargan and megalithic stone-box (*ashchechnye*) cemeteries across the Baidar valley and Southern Coast (Malorechenskoye, Oreanda, Simeiz, Gurzuf), the Tauri and pre-colonial steppe communities sustained a regional population of 48,0k (48 000) at 1000BC (Sub-region I: 17,5k; Sub-region II: 7,00k; Sub-region III: 13,0k; Sub-region IV: 10,5k).

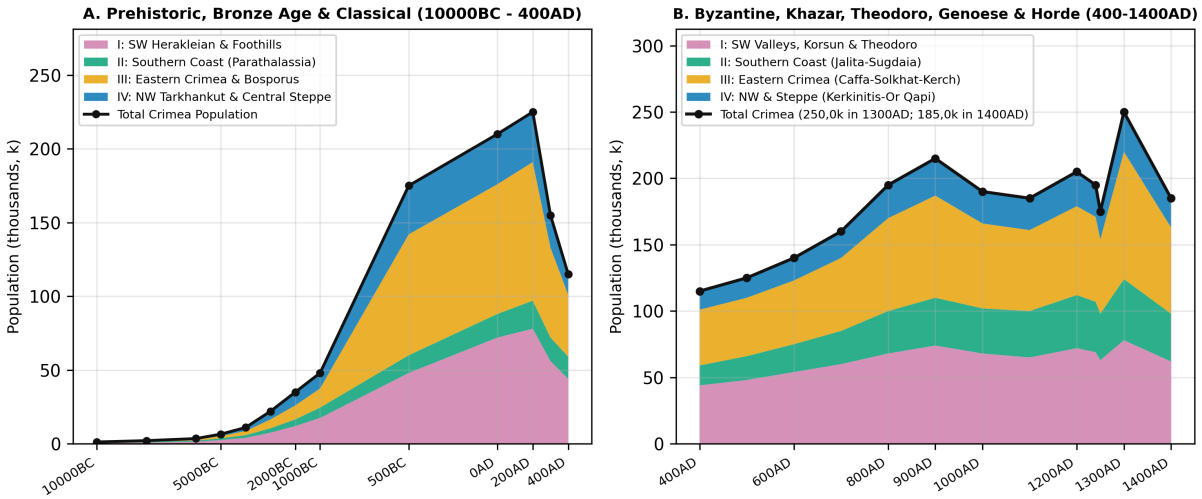


Figure 2. Demographic Trajectory of Crimea across Four Natural Sub-Regions (10000BC - 1400AD).

Classical Antiquity: Greek Poleis, Bosporan Chora and the Late Scythian Kingdom (500BC - 400AD)

The European Bosporan Kingdom and the Chersonesean Chora (500BC - 0AD)

Greek maritime colonisation in the 6th and 5th centuries BC triggered the first great demographic expansion in Crimean history, lifting total population from 48,0k in 1000BC to 175,0k in 500BC and 210,0k at 0AD [10][11][12]. Two distinct colonial systems reshaped the peninsula:

1. **The European Bosporus (Sub-region III, 82,0k at 500BC; 88,0k at 0AD; 94,0k at 200AD)**: Founded by Milesian colonists on Mount Mithridates c. 600BC, **Panticapaeum** (modern Kerch) became the capital of the Archaeanactid (480BC) and Spartocid (438-108BC) dynasties of the Bosporan Kingdom. At its Classical-Hellenistic and early Roman zenith, Panticapaeum enclosed ~100 ha of terraced acropolis, agora, craft quarters and harbour suburbs housing 22,0k to 28,0k inhabitants [10][13]. Flanking the capital along the Kerch Strait and Azov-Black Sea coasts stood a dense urban network: **Theodosia** (annexed by Leucon I c. 380BC as a major grain-exporting port capable of berthing 100 ships, 6,0k at 500BC), **Nymphaeum** (4,5k), **Myrmekion** (3,5k), **Tyritake** (3,2k - 4,0k, famed for its fish-salting vats), **Kimmerikon** (2,5k - 3,0k), **Kytaia** (2,2k - 2,9k), **Porthmion** (1,8k), **Akra** (1,5k - 1,9k) and **Zenon's Chersonesus** (1,4k - 2,1k). Shielded behind the monumental earthen **Tyritake** and **Uzunlarsky Ramparts** (36 km long from the Sea of Azov to Lake Uzunlar), the Kerch interior supported over 280 rural estates and fortified towns (Andreevka-Yuzhnaya, Generalskoye-Zapadnoye, Polyanka, Zolotoe, Tasunovo, Mikhailovka, Artesian and Ilurat), supplying Athens with 400 000 *medimnoi* (~16 500 tonnes) of wheat annually under Leucon I [10][14].
2. **Dorian Chersonesus Taurica and the North-Western Crimean Chora (Sub-regions I & IV, 48,0k & 33,0k at 500BC)**: Founded by Heraclea Pontica in 422/421BC on Quarantine Bay (following an earlier 6th-century Ionian emporium), **Chersonesus Taurica** expanded across a 38 ha Orthogonal grid enclosed by ashlar walls, supporting 4,5k inhabitants in 500BC and rising to 16,0k in 0AD and 20,0k in 200AD [15][16]. On the Herakleian Peninsula (~10 000 ha), G. M. Nikolaenko and J. C. Carter's surveys have mapped 415 stone-walled cadastral land-lots (*kleroi* of 26,5 ha each, subdivided into vineyards and orchards with fortified *pyrgoi* farmhouse towers), supporting an immediate suburban *chora* population of 6,0k - 8,0k [12][17]. In the mid-4th century BC, Chersonesus incorporated the **North-Western Crimean Chora** along the Tarkhankut coast, including the older Ionian polis of **Kerkinitis** (Yevpatoria, 8,5 ha, 4,2k at 500BC), **Kalos Limen** (Chornomorske, 2,5k - 2,8k) and dozens of fortified coastal estates and grain-collecting stations at **Panskoye I**, **Belyaus**, **Chaika**, **Masliny**, **Kara-Tobe**, **Kulchuk**, **Tarpanchi** and **Dzhangul** [11][18][19].

The Late Scythian Kingdom, Mithridates VI and the Roman Military Protectorate (200BC - 400AD)

Under pressure from Sarmatian tribes crossing the Don in the 3rd and 2nd centuries BC, Royal Scythian elites consolidated the **Late Scythian State in Crimea** (*Scythia Minor*), establishing their fortified capital at **Scythian Neapolis** (*Kermen-Chuk*, 20 ha stone-walled citadel on the Petrovskaya rocks in modern Simferopol, housing King Skilurus's mausoleum, megaron palace and ashlar granaries, with 7,5k - 8,0k inhabitants in 0AD - 200AD) [20][21][22]. Late Scythian kings Skilurus and Palakus overran the North-Western Chersonesean *chora* (occupying Kerkinitis, Kalos Limen, Chaika, Belyaus and Tarpanchi) and constructed a chain of over 50 fortified *gorodishcha* along the Alma, Kacha, Belbek and Salgir foothill rivers, led by the coastal port-citadel of **Ust-Alma** (Palakion, 6 ha, 3,5k - 4,0k), **Kermen-Kyr** (Chabion, 2,8k), **Bulganak** (Khaboi, 2,5k), **Alma-Kermen** (2,2k), **Belogorsk-Ak-Kaya** (2,8k), **Zayachye** (1,5k) and **Dobroye** (1,6k) [18][21][23].

Intervention by Mithridates VI Eupator's general Diophantus (110-107BC) rescued Chersonesus and integrated Crimea into the Pontic Empire before Roman hegemony was established after 63BC [16]. Under

the Principate (1st - 3rd centuries AD), Rome stationed vexillations of Legions *I Italica*, *V Macedonica* and *XI Claudia* along with the *Classis Flavia Moesica* at the Chersonesus citadel, Balaklava-Kadykovka (Symbolon) and the cliff-top fortress of **Charax** on Cape Ai-Todor (Sub-region II, 1,8k troops and *canabae* residents at 200AD) [16][24]. Simultaneously, the Bosporan Kingdom under the Sarmatian-descended Tibulii-Sauromates dynasty constructed heavy stone fortresses across the Kerch interior to guard against steppe incursions (**Ilurat** 2,6k, **Artesian** 2,1k, **Mikhailovka** 1,8k, **Tasunovo** 1,3k), bringing Crimea to its classical demographic apogee of **225,0k (225 000) in 200AD** (Sub-region I: 78,0k; Sub-region II: 19,0k; Sub-region III: 94,0k; Sub-region IV: 34,0k) [10][13].

The Gothic and Borani naval raids of 252-275AD (which sacked Tanais, Gorgippia, Ilurat, Myrmekion and the Late Scythian foothill citadels) and the subsequent **Hunnic invasion across the Kerch Strait in 370-376AD** (which destroyed the Bosporan rural *chora*, Scythian Neapolis and Ust-Alma) triggered a severe Late Antique demographic trough [23][25]. Regional population contracted to 155,0k in 300AD and bottomed out at **115,0k in 400AD** (-48,9% below the 200AD peak), as survivors retreated behind the walls of Chersonesus and Bosporos or migrated into the forested valleys of the Crimean Mountains (where the Alanic-Gothic cemeteries of Neyzats, Opushki, Druzhnoye, Suuk-Su and Luchistoye mark the genesis of medieval Crimean Gothia) [25].

Medieval Crimea: Byzantine Cherson, Gothia, Khazar Tudunate, Kievan Rus, Golden Horde and Genoese Gazaria (500AD - 1400AD)

Early Byzantine Reconquest, the Cave-Towns of Gothia and the Khazar / Saltovo-Mayaki Boom (500AD - 900AD)

Under Emperor Justinian I (527-565AD) and Maurice (582-602AD), Byzantium refortified Chersonesus (*Cherson*) and Bosporos, erected the coastal castles of **Aluston** (Alushta) and **Gorzouvitai** (Gurzuf) and constructed the *Long Walls* closing the mountain passes around the Gothic-Alanic federate territory of *Dory* [25][26]. Across the Inner Cuesta of Sub-region I, Byzantine military engineers and local Christianised Gothic-Alanic communities founded the monumental cliff-top "cave-towns" (*peshchernye goroda*): **Mangup-Doros** (the 90 ha mesa plateau capital, 2,5k in 500AD rising to 7,0k in 900AD), **Eski-Kermen** (18 ha plateau with a three-aisled basilica and rock-cut casemates, 2,8k in 600AD to 5,2k in 900AD), **Chufut-Kale** (*Qirq-Yer*, 3,2k in 900AD), **Bakla** (3,4k in 900AD), **Tepe-Kermen** (2,4k), **Kachi-Kalyon** (2,1k), **Kyz-Kermen** (1,8k in 800AD) and **Kalamita** (Inkerman, 2,0k) [26][27][28]. Regional population recovered steadily from 125,0k in 500AD to 140,0k in 600AD and 160,0k in 700AD.

During the 8th and 9th centuries AD, the establishment of the **Khazar Khaganate** and its Crimean governorate (administered by the Khazar *tudun* at Bosporos, Sugdaia and periodically Cherson) ushered in the **Saltovo-Mayaki demographic expansion** documented by I. A. Baranov and A. I. Aibabin [29][30]. Semi-nomadic Proto-Bulgarian and Alanic pastoralists sedentarised across over 260 open villages, potteries and vineyards in Eastern and Foothill Crimea:

- **Sugdaia** (Sudak, Sub-region II) emerged as a major trans-Pontic emporium (7,5k in 800AD; 9,0k in 900AD), flanked by the sprawling 18 ha coastal craft and monastic centre of **Tepsen** (Koktebel, 3,4k in 900AD), **Kordon-Oba** (1,8k), **Otuz** (Shchebetovka, 1,4k), **Geroevskoye** and **Ptashkino** on the Kerch Peninsula [29][31].
- In **Byzantine Cherson** (elevated to the capital of the *Theme of Klimata / Cherson* under Emperor Theophilos in 833/841AD), urban population reached 11,5k - 12,0k in 800-900AD [32][33]. Total Crimean population climbed to **195,0k in 800AD** and attained an early medieval peak of **215,0k in 900AD** (Sub-region I: 74,0k; Sub-region II: 36,0k; Sub-region III: 77,0k; Sub-region IV: 28,0k).

Kievan Rus Korsun and Korchev, Pecheneg-Cuman Steppe Raids and the 1200AD Recovery (900AD - 1250AD)

The arrival of the Pechenegs in the Pontic steppes (c. 900-960AD) and Prince Svyatoslav Igorevich's destruction of the Khazar Khaganate (965AD) disrupted open Saltovo-Mayaki villages in Eastern Crimea and the Kerch steppe (including the burning of Tepsen and Kordon-Oba), causing regional population to contract to 190,0k in 1000AD and 185,0k in 1100AD [29][34]. Simultaneously, Crimea was drawn directly into the political and ecclesiastical orbit of **Kievan Rus**:

1. **The Korsun Campaign of Vladimir the Great (988-989AD)**: Following a multi-month siege that severed the city's water pipes, Grand Prince Vladimir Svyatoslavich captured Byzantine Cherson (*Korsun*), accepted baptism in the city's episcopal complex, married the Porphyrogenita princess Anna and returned Korsun to Emperor Basil II as his bride-gift while transferring priests, icons and bronze quadriga statues to Kiev [32][33]. Throughout the 11th and 12th centuries, Korsun (10,0k in 1000AD; 8,0k - 8,5k in 1100-1200AD) functioned as the premier Byzantine-Rus trade staple (*Korsuniane* merchants in Kiev and Novgorod).
2. **The Rus Principality of Tmutarakan and Korchev (988-1094AD)**: Across the Kerch Strait, the Rurikid princes of Tmutarakan (Mstislav Vladimirovich, Gleb Svyatoslavich, Rostislav Vladimirovich and Oleg Svyatoslavich) ruled **Korchev** (ancient Bosporos / Kerch, 7,5k in 1000AD; 6,5k - 7,0k in 1100-1200AD), as commemorated by Prince Gleb's famous 1068AD *Tmutarakan Stone* measuring the 14 000-fathom ice crossing between Tmutarakan and Korchev [34].

By 1200AD, under the stable commercial condominium of the Polovtsians (Cumans), the Empire of Trebizond and the Seljuk Sultanate of Rum, **Sugdaia** (*Surozh* in Russian chronicles) reached 11,0k inhabitants as the chief silk-and-fur port of the Black Sea, while the fortified mountain towns of Gothia (Mangup 7,0k, Eski-Kermen 4,8k, Chufut-Kale 3,8k, Bakla 3,0k, Aluston 3,5k, Gorzouvitai 2,8k) expanded, lifting total Crimean population to **205,0k in 1200AD** [28][33].

The Mongol Invasions, Golden Horde Solkhat, Genoese Gazaria and the Principality of Theodoro (1223AD - 1400AD)

Mongol armies under Jebe and Subutai raided Sugdaia in January 1223AD, followed by Batu's conquest of the Crimean steppe in 1238-1239AD and Nogai's devastating punitive campaign of 1299AD (which sacked Korsun, Eski-Kermen, Bakla and Sugdaia to avenge the killing of his grandson Aktaji) [33][35]. While the initial Mongol shock reduced Crimean population from 195,0k in 1240AD to a brief trough of 175,0k in 1250AD, the integration of Crimea into the trans-Eurasian *Pax Mongolica* as the western maritime gateway of the **Golden Horde** (*Ulus of Jochi*) generated an unprecedented urban and commercial boom by **1300AD (250,0k regional total)**:

- **Golden Horde Solkhat / Qirim (Staryi Krym)**: Established in the sheltered Agarmysh valley of Sub-region III as the seat of the Golden Horde *tumen-bashi* (governor of the Crimean *ulus*), **Solkhat** (*Qirim*, from which the entire peninsula took its name) grew into a 300 ha walled Islamic and multi-confessional metropolis featuring the Mosque of Sultan Baybars (1287-1288AD), the Mosque and Madrasa of Uzbek Khan (1314AD), Armenian and Greek churches, caravanserais, mints and hydraulic pipe networks [35][36][37]. At its 1300-1340AD zenith, Solkhat housed **28,0k (28 000)** inhabitants (18,0k in 1400AD), complemented by the rising inland market city of **Karasubazar** (Bilohirsk, 8,0k in 1300AD; 7,5k in 1400AD) and the fortified cliff-top residence at **Chufut-Kale / Qirq-Yer** and **Eski-Yurt / Salachik** (Bakhchisaray valley, 5,5k and 3,5k in 1300-1400AD) [38][39][40][41].

- **Genoese Gazaria (Caffa, Soldaia, Cembalo and the Capitanate of Gotthia)**: Granted commercial concessions at ancient Theodosia by Mengu-Timur Khan in 1266AD, the Republic of Genoa transformed **Caffa** (Feodosia) into a massive double-walled maritime city enclosing 90 ha (citadel and outer *burgh* with 26 towers protecting 11 000 - 15 000 urban plots and suburban *borghi* populated by Armenians, Greeks, Italians, Tatars, Jews and Rus) [37][42][43][44]. Prior to the outbreak of the **Black Death** during Janibeg Khan's siege of Caffa in 1346AD, Caffa housed **32,0k - 40,0k** inhabitants at the 1300-1345AD peak (Sub-region III total: 96,0k in 1300AD). Genoa further acquired **Cembalo** (Balaklava, 1343/1357AD, 2,2k in 1300AD; 2,5k in 1400AD), **Soldaia** (Sudak, 1365AD, with its 18 coastal villages and monumental 14-tower fortress, 12,0k in 1300AD; 6,5k in 1400AD), **Vosporo** (Kerch, 5,5k in 1300AD; 4,5k in 1400AD) and the coastal castles of the *Capitanate of Gotthia* (Aluston / *Lusta* 4,0k, Gorzouvitai 3,2k, Partenit 2,6k, Jalita / Yalta 3,0k, Foros 1,6k) [28][31][44][45].
- **The Christian Principality of Theodoro (Mangup)**: In the southwestern mountains and Belbek-Kacha-Chorna valleys (Sub-region I), the Greek-Gothic dynasty of the Gabrades consolidated the **Principality of Theodoro**, expanding the citadel and three-line defensive walls of **Mangup** (8,5k in 1300AD; **9,0k at the 1400AD benchmark**), constructing the port fortress of **Kalamita** (Inkerman, 2,5k in 1400AD) to rival Genoese Cembalo, and fortifying the eastern border castle of **Funa** at the foot of Mount Demerdzhi alongside **Syuyren**, **Kermenchik** and **Chorgun** [27][28][35].

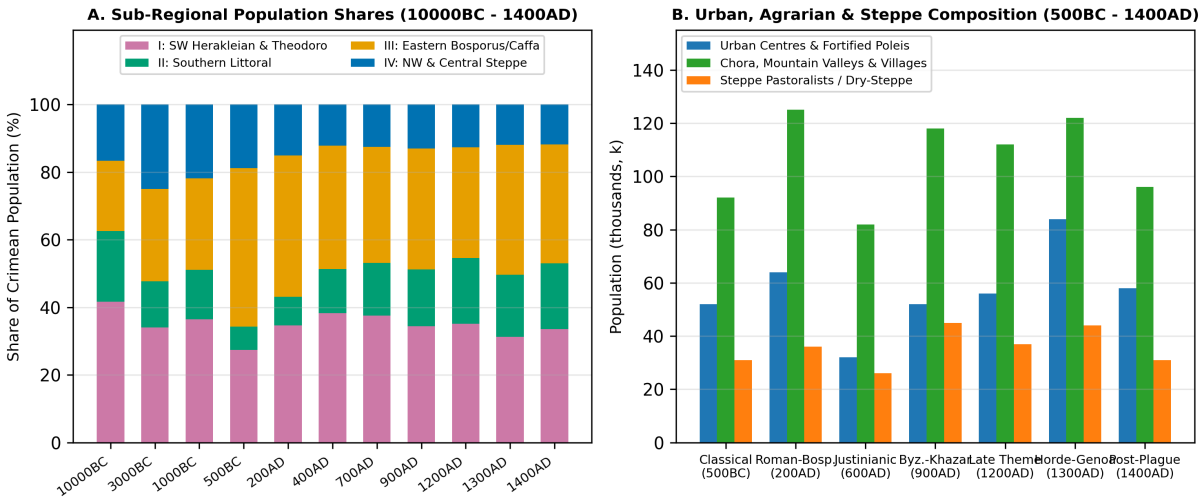


Figure 3. Sub-Regional Population Shares and Urban-Agrarian-Steppe Composition across Pre-1400AD Historical Horizons.

The 1346AD Black Death, Golden Horde Civil Wars and the 1400AD Cutoff

Recurrent waves of bubonic plague after Janibeg's 1346AD siege of Caffa, internecine warfare within the Golden Horde during the *Great Troubles* (*Velikaya Zamolnya*, 1359-1380AD, when Mamai controlled the Crimean *ulus*), and the campaigns of Timur (1395-1396AD) and Edigu (who besieged Korsun and sacked its remaining quarters in 1399AD) reduced Crimea's population from 250,0k in 1300AD to **185,0k at the 1400AD terminal benchmark** (Sub-region I: 62,0k; Sub-region II: 36,0k; Sub-region III: 65,0k; Sub-region IV: 22,0k; with urban populations at Caffa 22,0k, Solkhat 18,0k, Mangup 9,0k, Karasubazar 7,5k, Soldaia 6,5k, Qirq-Yer 5,5k, Vosporo 4,5k, Cembalo 2,5k and Korsun 2,0k) [35][42][46][47]. Because the subsequent secession of the independent Crimean Khanate under Haji I Giray (1441AD) and the Ottoman conquest of Caffa, Soldaia and Mangup (1475AD) initiated a distinct early modern fiscal and demographic regime, this literature review and its companion dasymetric calibration terminate at 1400AD.

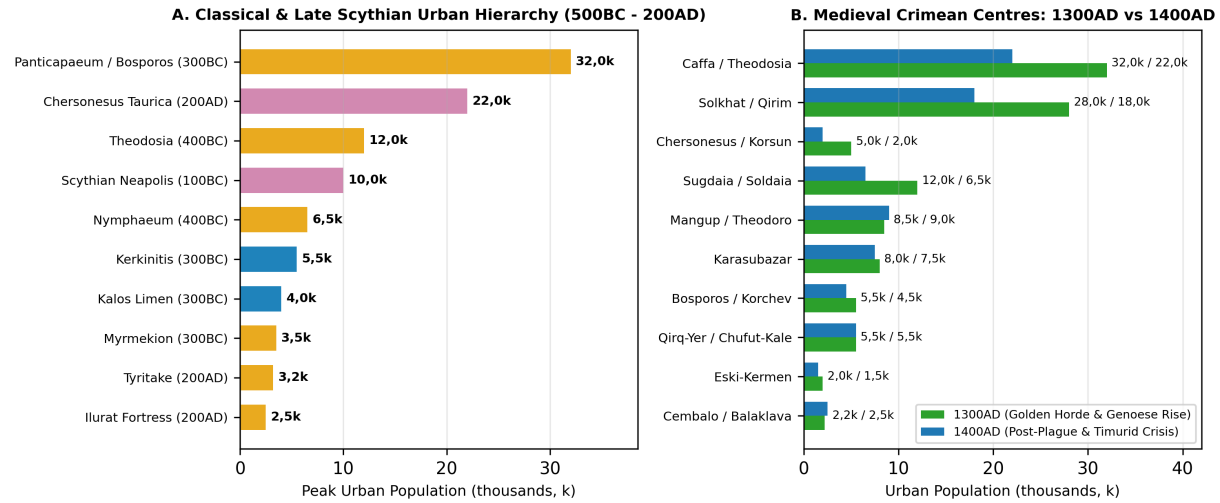


Figure 4. Urban Hierarchy of Crimea: Classical and Late Scythian Peaks (500BC - 200AD) versus Medieval Golden Horde, Genoese and Theodorite Centres (1300AD vs 1400AD).

Master Regional Population Estimates (10000BC - 1400AD)

The table below synthesises the complete benchmark population trajectory for Crimea (~26 945 km^2) and its four natural sub-regions from 10000BC to 1400AD.

Year	I: SW & Theodoro	II: South Coast	III: East & Bosporus	IV: NW & Steppe	Total Pop.	Historical & Archaeological Context
10000BC	0,50k	0,25k	0,25k	0,20k	1,2k	Upper Palaeolithic / Early Shan-Koba Grottoes (Kiik-Koba, Siuren I, Buran-Kaya III)
8000BC	0,85k	0,45k	0,45k	0,35k	2,1k	Boreal Shan-Koba Mesolithic Foragers (Shan-Koba, Fatma-Koba, Grot Skalisty)
6000BC	1,40k	0,75k	0,75k	0,60k	3,5k	Atlantic Murzak-Koba Mesolithic & Early Neolithic (Murzak-Koba, Laspi VII, Zamil-Koba)
5000BC	2,50k	1,20k	1,50k	1,30k	6,5k	Crimean Neolithic Expansion (Tash-Air I, Kaya-Arasy, Frontovoye I, At-Bash)
4000BC	4,00k	1,80k	2,80k	2,40k	11,0k	Eneolithic Steppe & Foothill Pastoralism (Sredny Stog & Early Copper Age)
3000BC	7,50k	3,00k	6,00k	5,50k	22,0k	Early Bronze Age Kemi-Oba Megalithic & Yamnaya Cultures (Kemi-Oba, Bakhchi-Eli)

(Continued from previous page)

Year	I: SW & Theodoro	II: South Coast	III: East & Bosporus	IV: NW & Steppe	Total Pop.	Historical & Archaeological Context
2000BC	12,0k	4,50k	9,50k	9,00k	35,0k	Middle Bronze Age Catacomb & Late Kemi-Oba Settlements (Kamenka, Planerskoye)
1000BC	17,5k	7,00k	13,0k	10,5k	48,0k	Late Bronze / Early Iron Age Tauri Kizil-Koba Culture (Uch-Bash, Kizil-Koba)
500BC	48,0k	12,0k	82,0k	33,0k	175,0k	Archaic-Classical Greek Colonisation (Panticapaeum, Chersonesus, Theodosia, Kerkinitis)
0AD	72,0k	16,0k	88,0k	34,0k	210,0k	Hellenistic-Roman Transition & Late Scythian State (Scythian Neapolis, Ust-Alma)
200AD	78,0k	19,0k	94,0k	34,0k	225,0k	Roman Protectorate & Bosporan Apogee (Charax, Chersonesus 20k, Panticapaeum 28k)
300AD	56,0k	16,0k	61,0k	22,0k	155,0k	Post-Gothic Raid Contraction (Destruction of Tanais, Ilurat & Scythian Foothill Forts)
400AD	44,0k	15,0k	42,0k	14,0k	115,0k	Post-Hunnic Trough (370-376AD Collapse of Bosporan Chora & Migration into Mountains)
500AD	48,0k	18,0k	44,0k	15,0k	125,0k	Early Byzantine Reconquest under Anastasius & Justinian I (Aluston, Gorzouvitai, Doros)
600AD	54,0k	21,0k	48,0k	17,0k	140,0k	Byzantine Exarchate & Rise of Gothic Cave-Towns (Mangup, Eski-Kermen, Chufut-Kale)
700AD	60,0k	25,0k	55,0k	20,0k	160,0k	Early Khazar Condominium & Crimean Tudunate (Cherson, Bosporos, Sugdaia)

(Continued from previous page)

Year	I: SW & Theodoro	II: South Coast	III: East & Bosporus	IV: NW & Steppe	Total Pop.	Historical & Archaeological Context
800AD	68,0k	32,0k	70,0k	25,0k	195,0k	Saltovo-Mayaki Sedentarisation Boom (260+ Eastern & Foothill Villages, Tepsen)
900AD	74,0k	36,0k	77,0k	28,0k	215,0k	Byzantine Theme of Klimata / Cherson & Khazar Apogee (Cherson 12k, Sugdaia 9k)
1000AD	68,0k	34,0k	64,0k	24,0k	190,0k	Kievan Rus Korsun Campaign (988AD) & Tmutarakan-Korchev Principality (Post-Khazar Shift)
1100AD	65,0k	35,0k	61,0k	24,0k	185,0k	Cuman (Polovtsian) Steppe Hegemony & Byzantine-Rus Trade at Korsun and Korchev
1200AD	72,0k	40,0k	67,0k	26,0k	205,0k	Pre-Mongol Commercial Recovery (Sugdaia/ Surozh 11k, Mangup 7k, Korsun 8k)
1240AD	69,0k	38,0k	64,0k	24,0k	195,0k	Initial Mongol Incursions (1223AD & 1238-1239AD Steppe Campaigns)
1250AD	63,0k	35,0k	56,0k	21,0k	175,0k	Post-Conquest Transition & Early Ulus of Jochi Reorganisation
1300AD	78,0k	46,0k	96,0k	30,0k	250,0k	Golden Horde Solkhat (28k) & Genoese Caffa (32k) Maritime Apogee (despite 1299AD Raid)
1400AD	62,0k	36,0k	65,0k	22,0k	185,0k	Post-1346AD Black Death, Horde Civil Wars & 1395-1399AD Timur/ Edigu Raids; Theodoro Peak

Table 2. Master Benchmark Population Estimates for Crimea by Natural Sub-Region (10000BC - 1400AD).

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